

IT Migration & Dependencies Brief

Nevada County Experience — technical reference for county IT staff

This document supports the technical evaluation of moving the Nevada County Experience platform onto county-owned infrastructure. It enumerates every external service the platform touches, the data flow in both directions, four hosting scenarios with cost and policy implications, a step-by-step migration plan, ongoing administrative responsibilities, and a policy-block impact matrix.

Key facts up front:

- Platform is a static HTML/JS site plus a small Python (Flask) admin server.
- No database — all data lives in JSON files inside the project directory.
- No visitor accounts, no PII collected; itinerary save is opt-in localStorage.
- Outbound network calls only — no inbound listening except HTTPS/443.
- AI categorization is optional; site is fully functional without it.
- Total monthly cost on a \$6/mo VPS: ~\$7-8 including AI use at chamber scale.

1. External Services Inventory

Every external dependency the platform touches, with vendor, purpose, data flow direction, monthly cost, what fails if the service is blocked or down, and replacement options. **Outbound** = our server initiates; **Inbound** = visitor reaches us. There are no inbound vendor connections.

Service	Purpose	Direction	Cost	Required?
Anthropic Claude API	AI categorize button — refines tags, venue, area, quality flags	Outbound HTTPS to api.anthropic.com	\$0.30–\$1/mo	Optional
Plausible Analytics	Privacy-respecting aggregate metrics (page views, itinerary, etc.)	Outbound HTTP to plausible.io	\$0 (self-host) or \$9/mo	Optional, opt-in
GitHub Pages	Static-site hosting (current public deploy). Replaceable by any web server	Outbound HTTPS to github.com on deploy	\$0 only	Replaceable
Google Fonts CDN	Playfair Display + Josefin Sans typefaces. Hotlinked from fonts.gstatic.com	Outbound HTTPS from visitor browser to fonts.googleapis.com	\$0	Replaceable (self-host)
Leaflet (Cloudflare CDN)	Map library for the "View on Map" feature. Hotlinked from cdn.jsdelivr.net	Outbound HTTPS from visitor browser	\$0	Replaceable (self-host)
Unsplash CDN	Stock photography for theme cards. ~30 hotlinked images	Outbound HTTPS from visitor browser to images.unsplash.com	\$0	Replaceable (self-host)
KVMR website (scrape target)	Community-radio events calendar — major source of weekly events	Outbound HTTPS from server (kvmr.org)	\$0	Optional source

Service	Purpose	Direction	Cost	Required?
Trumba JSON feed (NCAC)	Nevada County Arts Council calendar — 540 events, 16-month window. JSON, trumba.com	Outbound HTTP to trumba.com	\$0	Optional source
Eventbrite, NC Chamber, GV Chamber, The Olive	Arts and theater event scrape sources. Public HTML pages, no auth	Outbound HTTPS from server	\$0	Optional sources
Let's Encrypt	Free TLS certificates via certbot. Required for HTTPS	Outbound HTTPS to acme-v02.api.letsencrypt.org during cert renewal	\$0	Required for HTTPS
Gmail SMTP (optional alerting)	Currently used by trade-related projects; not used by this platform	Outbound	N/A	Not used

What we do not use: no third-party tracking / advertising / identity / payment / CDN-for-data services. No social-login providers. No vendor SDK or proprietary library — the entire site is plain HTML/JS that opens in any modern browser.

2. Architecture & Data Flow

2a. Components

Component	Tech	Disk path	Network
Public static site	HTML + JS (single file)	index.html	Served on :443 to visitors
Admin server	Python 3.12 + Flask	server.py	Listens on :5000 behind nginx
Event scraper	Python 3 + Selenium (Chromium headless)	scraper/event_scraper.py	Outbound HTTPS to source sites
AI categorizer	Python 3 + Anthropic SDK	scraper/ai_categorize.py	Outbound HTTPS to api.anthropic.com
Events queue file	JSON	scraper_output/events.json	Local only
Suggestions file	JSON	scraper_output/suggestions.json	Local only
Scraper sources file	JSON	scraper/sources.json	Local only
Itinerary state	localStorage in visitor's browser	n/a (client-side)	Never leaves the visitor's device

2b. Data flow

Visitor path: browser → CDN (Google Fonts, Leaflet, Unsplash, optional Plausible) and HTTPS to our server :443 → nginx reverse-proxies to Flask on :5000 → returns index.html and JSON event data. No cookies set; no server-side session.

Admin path: chamber staff browser → HTTPS to /admin (gated by either localhost-only access or ?admin=1 query parameter on a whitelisted hostname) → Flask serves the admin endpoints (events queue, suggestions, scraper trigger, AI categorize trigger).

Scheduled job path: cron / systemd timer / Windows Task Scheduler → invokes Python scraper → scraper fetches HTML/JSON from each enabled source → writes new entries to events.json → optionally invokes AI categorizer which posts to Anthropic API and writes enriched fields back to events.json.

What flows OUT of the county: only the scrape requests (public-page fetches), AI categorize prompts (event titles + descriptions, no PII), and Let's Encrypt cert renewal calls. No visitor data is ever transmitted to a third party.

3. Hosting Scenarios

Four realistic options for where the platform runs. Pick by whatever best matches county policy on cloud vs. on-prem and OS preference. Migration effort is similar across A, B, and C (Linux). Scenario D (Windows Server) is the heaviest lift because the scraper expects a POSIX environment.

3a. Scenario A — Linux VPS (DigitalOcean, Vultr, Linode)

Profile: \$6–12/month managed VPS. Ubuntu 24.04 LTS. We provision, county pays the bill directly. County retains root SSH access; vendor handles hypervisor and provider backups.

Effort: half-day for a Linux-comfortable admin; two days for a learner. **Monthly:** \$6 (droplet) + \$1 (provider backups) + ~\$1 (AI) = \$8/mo. **Best for:** chambers that prefer "we know one person who can SSH in and fix things" over an in-house ops team.

3b. Scenario B — AWS / Azure / GCP IaaS

Profile: AWS EC2 t4g.small or Azure Standard_B1ms, Ubuntu image. Roughly \$15–25/month for the smallest instance the platform needs. County may already have an enterprise account, security baselines, and an existing landing-zone configuration to drop into.

Effort: half-day if the county has paved-road provisioning; one day otherwise. **Monthly:** \$15–25 (instance) + \$5 (EBS storage & snapshots) + ~\$1 (AI) = \$21–31/mo. **Best for:** counties standardized on a public-cloud vendor with existing tooling.

3c. Scenario C — County on-prem VMware / Hyper-V (Linux VM)

Profile: Ubuntu 24.04 LTS as a guest on county-managed hypervisor. 2 vCPU, 4 GB RAM, 30 GB disk is plenty. County handles host-level backups, patching cadence, and network access; we install the application stack inside the VM.

Effort: half-day to a day depending on how long VM provisioning takes through the county's request process. **Monthly:** incremental (counted as part of existing virtualization licensing). AI use still ~\$1/mo if outbound is permitted. **Best for:** counties with existing private-cloud capacity and policy preference for on-prem.

3d. Scenario D — Windows Server (IIS or Python service)

Profile: Windows Server 2022, IIS reverse-proxies to the Flask admin server, scraper runs as a Windows Service via NSSM or as a scheduled task via Task Scheduler. Chromium must be installed for Selenium; HTTPS via win-acme (alternative to certbot).

Effort: one to two days. The scraper depends on Selenium with headless Chromium — works on Windows but is less battle-tested than Linux. Chocolatey or winget for installs.

Monthly: incremental (existing Windows Server licensing) + AI ~\$1. **Best for:** counties whose IT shop is exclusively Windows and has no Linux administration capacity.

Sizing summary (all scenarios):

Resource	Minimum	Recommended	Why
CPU	1 vCPU	2 vCPU	Headless Chromium during scrape; 1-2 minute peak
RAM	2 GB	4 GB	Chromium memory footprint; Flask itself is tiny
Disk	10 GB	30 GB	Code under 100 MB; JSON files grow slowly; backups optional
Bandwidth	5 GB/mo egress	20 GB/mo	Static site; map/font CDN traffic served by 3rd parties
Inbound ports	443 only	443 + 22 (SSH limited)	80→443 redirect; SSH locked to admin IPs

4. Migration Plan

Ten phases, each independently verifiable. The walkthrough assumes Linux (scenarios A/B/C); Windows differences are called out where relevant. A Linux-comfortable admin completes phases 0–9 in a focused half-day.

Phase	What happens	Owner	Verifiable when
0. Decisions	Pick hosting scenario, domain name, DNS provider, who owns the bill. Generate Anthropic API key (optional)	County	Hosting account + domain registered
1. Pre-flight	Domain DNS access confirmed, SSH key pair created and saved, Anthropic credentials in password manager	County	AI credentials in a shared password vault
2. Provision	VM/droplet created, Ubuntu 24.04 LTS installed, security baseline applied (firewall, fail2ban, automatic updates), <code>sudo apt install python3-pip</code>	County	<code>ssh@3.12.7</code> connects, <code>uname -a</code> installed
3. Deploy	Repository cloned to <code>/opt/ncexp</code> , Python venv created, requirements.txt installed, <code>python server.py</code> starts cleanly on port 5000	County	<code>python server.py</code> starts cleanly on port 5000
4. systemd service	<code>ncexp.service</code> unit file installed under <code>/etc/systemd/system</code> , enabled to auto-start on boot, auto-restart on crash	County	<code>systemctl status ncexp</code> reports active (running)
5. nginx + TLS	nginx vhost reverse-proxies localhost:5000 to public 443. Certbot issues Let's Encrypt certificate; auto-renewal installed	County	<code>curl https://experience.county.gov</code> returns the site
6. Schedule scrapes	cron entries (Linux) or Task Scheduler entries (Windows) added: nightly source refresh at 3 AM, AI categorization at 3:30 AM, daily backups at 4 AM	County	Successful 3 AM cron job
7. Backups	Provider snapshots enabled (DO/AWS) OR county VM backup policy applied (COW)	County	Daily JSON tarball mirror restored (SQL, yesterday's backup, verify events.js)
8. Test checklist	Public site loads in all major browsers; admin Events Queue accepts approved dismisses; AI Categorize button does not error in the test (sleep is disabled); /feed.	County	Test passes
9. Hand-off	Document handed to chamber: admin URL, log paths, common troubleshooting, escalation contact. County Chamber operator runs through the operator_guide.	County	Chamber operator

Phase 6 — schedule example (Linux cron)

```
# Edit the ncxp user's crontab
sudo -u ncxp crontab -e
```

```
# Add three lines (UTC times – adjust for PT if server isn't on PT):
0 3 * * * cd /opt/ncexp && .venv/bin/python scraper/event_scraper.py >> /var/log/ncexp-scrape.log 2>&1
30 3 * * * cd /opt/ncexp && .venv/bin/python scraper/ai_categorize.py >> /var/log/ncexp-ai.log 2>&1
0 4 * * * tar czf /var/backups/ncexp-$(date +%Y%m%d).tgz /opt/ncexp/scraper_output
```

Phase 6 — schedule example (Windows Task Scheduler)

```
schtasks /Create /TN "NCExp-Scrape" /TR "C:\ncexp\.venv\Scripts\python.exe C:\ncexp\scraper\event_scraper.py" /SC DAILY /ST 03:00 /RU ncexp
schtasks /Create /TN "NCExp-AI" /TR "C:\ncexp\.venv\Scripts\python.exe C:\ncexp\scraper\ai_categorize.py" /SC DAILY /ST 03:30 /RU ncexp
```

5. Administration Responsibilities & Frequencies

Two roles maintain the platform: **chamber operator** (content curation, queue review) and **county IT** (server health, security patches, backups). Time estimates assume the platform is running normally; incidents are separate.

5a. Chamber operator tasks

Task	Frequency	Time	Tool
Review the pending Events Queue, approve / dismiss	2–3× per week	5–10 min	Admin → Events Queue tab
Run AI Categorize after a scrape	After each scrape	~30 sec (click + wait)	Admin → AI Categorize button
Click ■ Prune Past between scrapes if queue grows	Ad-hoc	30 sec	Admin → Events Queue → ■ Prune Past
Review the Suggestions tab (visitor-submitted ideas)	Weekly	10–15 min	Admin → Suggestions tab
Spot-check that scrapers are not returning 0 events	Weekly	2 min	Admin → Events Queue status bar
Publish a Suggested Experience (curated itinerary)	As inspiration strikes	15–30 min	My Itinerary → ■ Publish
Edit / delete a published Suggested Experience	Ad-hoc	5 min	Suggested Experiences tab → ✎ Edit
Add a new venue / experience	When a member opens	5 min	Admin → Experiences tab → + Add row
Update an experience's hours, photo, or notes	Ad-hoc	2 min	Admin → Experiences tab → inline edit
Add a new event-source URL	When discovered	10 min	Admin → Event Sources tab
Quarterly review of disabled scraper sources	Quarterly	30 min	Admin → Event Sources, re-test blocked
Annual full-catalog walk-through (verify URLs, hours)	Annual	4–6 hours	Admin → Experiences, row by row

Estimated steady-state chamber operator time: ~30 minutes per week during normal operation, plus a 4–6 hour annual review.

5b. County IT tasks

Task	Frequency	Time	Tool / location
OS security updates	Weekly (auto, monitored)	5 min review	`unattended-upgrades` log
Application updates (git pull from upstream)	Monthly or as needed	15 min	`git pull && systemctl restart ncexp`
Backup verification — pull last tarball and dry-run restore	Monthly	15 min	`tar tzf ncexp-YYYYMMDD.tgz`
TLS certificate expiry check (certbot auto-renews; verify)	Quarterly	5 min	`certbot certificates`
Log rotation health	Quarterly	5 min	`ls -la /var/log/ncexp-*.log`

Task	Frequency	Time	Tool / location
Disk usage review	Quarterly	5 min	<code>`df -h` / `du -sh /opt/ncexp/scrapper_output`</code>
User access review — SSH keys, admin URLs	Semi-annual	15 min	<code>`~/.ssh/authorized_keys`</code> ; admin host whitelist
DR / restore drill — full server rebuild from backup	Annual	2–4 hours	Snapshot → fresh VM → restore JSON
Anthropic API key rotation	Annual or on staff change	10 min	Anthropic console + <code>`config.py`</code>
Vendor / external-service review (still permitted? alternatives?)	Annual	30 min	This document, section 1

Estimated steady-state county IT time: ~30 minutes per month during normal operation, plus an annual DR drill.

6. Security & Compliance Posture

Control	How it's handled
Inbound network exposure	HTTPS only (port 443). SSH (22) restricted to admin IPs via firewall. No other listening ports.
TLS / certificates	Let's Encrypt via certbot; auto-renew cron. Strong cipher suite per Mozilla "intermediate" profile.
Secrets storage	API keys and credentials in `config.py` (file mode 0600, owned by service user). Not in git. Optional: load from environment variables.
Authentication	Visitor: none required. Admin URL: localhost-only by default OR `?admin=1` query parameter on a whitelisted hostname. For public admin,
Authorization	Single chamber operator role; no fine-grained RBAC. Add nginx Basic auth users for multiple operators if required.
Personal data (PII)	None collected. Itinerary save is opt-in localStorage on the visitor device — never leaves their browser. Suggestion form collects only the op
GDPR / CCPA scope	Effectively out of scope: no cookies, no PII storage. Suggestion-form emails (if entered) fall under standard records-retention policy.
ADA / WCAG	WCAG 2.1 AA on the demo. High-contrast text, keyboard-navigable, semantic HTML. Re-audit recommended after each major UI change.
Logging	`/var/log/ncexp-scraper.log`, `/var/log/ncexp-ai.log`, `journalctl -u ncexp` for the web server. No visitor-IP retention by default.
Patch cadence	OS: `unattended-upgrades` for security patches. Application: monthly `git pull` review. Python deps: quarterly `pip list --outdated`.
Incident response	Site is stateless apart from the JSON files; full restore from backup is < 30 minutes. Provider snapshot rollback covers OS-level compromises
Supply chain	All Python deps pinned in requirements.txt. Frontend deps via CDN (Leaflet, fonts) — self-host option documented in section 7 if CDN use i

7. Flask Server — Security Review

The public site is plain HTML/JS and can be deployed to any web server with no special review. The **Flask admin server** is the only component that warrants security scrutiny — it executes Python, spawns subprocesses (headless Chromium for scrapes), holds API credentials, and writes to the local filesystem. This section enumerates the concerns and the mitigations available out of the box.

7a. What Flask does (in one paragraph)

A ~280-line Python script (server . py) using Flask. Listens on localhost : 5000 behind an nginx reverse proxy. Serves the static index . html; exposes a small JSON API used by the in-page admin tabs (Events Queue, AI Categorize trigger, Suggestions review, Scraper Sources). Stateless across restarts — every request reads/writes JSON files on disk; no database, no in-memory sessions.

7b. Concerns IT will raise (and pre-built mitigations)

Concern	Default state	Mitigation	Effort
Admin endpoints have no login wall	Gated by network only — listens on localhost	Add HTTP Basic authentication on /admin* paths in OpenAPI spec to SAML/OIDC v	12/18/2021
Endpoints execute subprocesses (Python / scraper / AI / categorize spawn child	Run Flask as an unprivileged user (oceap) with a restricted shell; subprocesses inherit the same low-		5 hours
Long-running Python process owns critical files (fig.py (mode 0600). Writes to scraper_output/	Add a .py file to fig.py (mode 0600). Writes to scraper_output/	Privileges=true, ProtectSystem=strict, ProtectHome=true, PrivateTmp=true	2 hours
Containerization / process isolation	Bare-metal Python process under systemd.	Optional: run in Docker/Podman with --read-only root, --cap-drop=ALL, --device-m	2 hours
Public suggestion form open to abuse	POST /api/suggestions accepts visitor input (Add email to description) in 30 seconds /	Appends suggestions to OptaCloudflare Turnstile (free, no	60 min
Python / Flask / Selenium / Chromium / pip packages pinned in requirements.txt. Chromium is pinned via weekly	Schedule pip update via weekly (or use Dependabot if mirrored to county	15 min	15 min
No audit trail of admin actions	Admin clicks (approve/dismiss) are appended to events . json	Log events . json to a separate file. Logged with readable identifier	15 min
What if Flask itself is compromised?	Worst-case: attacker has read/write on scraper_output/	Provides a snapshot of the Anthropic API. Anthropic key has a \$5/minute hard spending cap. No	\$5/minute

7c. The "split deployment" option for strictest policies

If county policy treats the dynamic Flask layer as too sensitive to host on its own server, the site can be split:

Public site	Static HTML/JS	Hosted on a CDN or county static-web bucket. Zero attack surface. No Python, no inbound
Admin layer	Flask + scraper	Hosted on a small internal VM the chamber operator reaches via VPN or county intranet only. Public internet never
Data hand-off	Generated artifact	Admin layer publishes its scraper_output/events.json to the public bucket on a cron (rsync, aws s3 cp, or equivalent)

Trade-off: the "Suggest a venue" public form needs SOME live endpoint — either keep a tiny suggestions-only Flask on a hardened sub-host, replace it with a static form-to-email service (Formspree, Tally, Microsoft Forms), or remove the public form entirely. Everything else works in the split model.

7d. What runs entirely without Flask

The visitor experience is HTML/JS only. With Flask off and only the static `index.html` + a periodically-updated `scraper_output/events.json` available, visitors get: theme browsing, vibe-pill filtering, day-by-day itinerary builder, opt-in `localStorage` save, share-link generation, map view, print, and Suggested Experiences browse. The only things that require Flask are the chamber-side admin tools and the public Suggest-a-venue submission form.

8. Policy-Block Impact Matrix

For each external service, what breaks if county policy disallows it and what the workaround is. Sites continues to function in nearly all cases; severity scales with how visitor-facing the lost feature is.

If this is blocked...	Impact	Workaround
Outbound HTTPS to api.anthropic.com	AI Categorize button stops working. Newly-scraped events keep hidden from AI Categorize button to tags admin, already one category review website is able to	Hide the AI Categorize button to tags admin, already one category review website is able to
Outbound HTTPS to plausible.io	Aggregate metrics (visitor counts, itinerary build rate) unavailable. Site is still Plausible, and the share of PS (operational vs. \$0 bill) is not affected. Or u	See the Plausible dashboard for share of PS (operational vs. \$0 bill) is not affected. Or u
Outbound HTTPS to fonts.googleapis.com (visitor-facing)	Custom fonts fall back to system serif/sans-serif. Site readable, slightly less "Paid Display	Slightly less "Paid Display + Josefin Sans WOFF2 files from /static/fonts
Outbound HTTPS to cdnjs.cloudflare.com (Leaflet)	Map view button fails. Itinerary still works; visitors lose the "View Self-Map" feature.	Self-Map" feature (one JS + one CSS file, ~150 KB total) from /static/leaflet/.
Outbound HTTPS to images.unsplash.com	~30 theme-card photos fail to load. Theme cards render with the original Unsplash placeholder, usable	Download Unsplash photos to site, upload to /static/photos/, update the THE
Outbound HTTPS to scraper source sites (KVM, Tumblr, etc.)	WP, Tumblr, etc. live. Existing events keep showing until they expire. Static catalog gets stale with 2 weeks	Operate the static catalog with 2 weeks events manually via the admin
Outbound HTTPS to acme-v02.api.letsencrypt.org (certificate renewal)	Let's Encrypt certificate eventually expires (90-day life). Site becomes inaccessible if not renewed	Use a certificate from CA, or a paid cert from a vendor the county already uses
Inbound HTTPS (443) from public internet	Site is internal-only. Visitors outside the county network can't reach it.	Repurpose as a kiosk / intranet tool. Defeats the public-facing visitor-planning
Cron / Task Scheduler permitted to run as a service on server	Scrape source run automatically. Queue stays empty. Auto-prune of past events	Operator manually checks for "Update Now" in the admin UI 1–3 times per week
Outbound SMTP (any provider)	No effect on this platform — we don't send email. Reserved for future use.	Future use only, yet built.

The "everything off but the lights" scenario: with every optional outbound service blocked (no Anthropic, no Plausible, no CDNs, no scrape, no Let's Encrypt), the platform still functions as a manually-curated static catalog hosted on the county server with a county-internal certificate. Operator time roughly doubles (~1 hour/week instead of 30 min) because manual tagging and manual event entry replace the automated paths. No visitor-facing feature is broken; only the data-freshness rate drops.

9. Appendices

A. systemd unit file

```
[Unit]
Description=Nevada County Experience admin server
After=network.target

[Service]
Type=simple
User=ncexp
WorkingDirectory=/opt/ncexp
Environment="ANTHROPIC_API_KEY=sk-ant-..."
ExecStart=/opt/ncexp/.venv/bin/python server.py
Restart=on-failure
RestartSec=5s

[Install]
WantedBy=multi-user.target
```

B. nginx vhost (HTTPS, with redirect from HTTP)

```
server {
    listen 80;
    server_name ncexperience.county.gov;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl http2;
    server_name ncexperience.county.gov;

    ssl_certificate      /etc/letsencrypt/live/ncexperience.county.gov/fullchain.pem;
    ssl_certificate_key  /etc/letsencrypt/live/ncexperience.county.gov/privkey.pem;
    include              /etc/letsencrypt/options-ssl-nginx.conf;

    # Strong default headers
    add_header Strict-Transport-Security "max-age=31536000; includeSubDomains" always;
    add_header X-Content-Type-Options    "nosniff" always;
    add_header X-Frame-Options           "SAMEORIGIN" always;

    location / {
        proxy_pass      http://127.0.0.1:5000;
        proxy_set_header Host          $host;
        proxy_set_header X-Real-IP     $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
```

```
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

C. Daily JSON backup script

```
#!/bin/bash
# /opt/ncexp/bin/backup-daily.sh – run by root cron at 04:00
set -euo pipefail
STAMP=$(date +%Y%m%d)
DEST=/var/backups/ncexp
mkdir -p "$DEST"
tar czf "$DEST/ncexp-$STAMP.tgz" /opt/ncexp/scrapper_output /opt/ncexp/config.py
# Retain 14 days
find "$DEST" -name 'ncexp-*.tgz' -mtime +14 -delete
# Optional: mirror to county SAN / S3
# rsync -a "$DEST/" backupserver:/backups/ncexp/
```

D. Reference URLs

Document / artifact	Where
Live demo (currently on GitHub Pages)	https://liammlrb-eng.github.io/nevada-county-experiences/
Source repository	https://github.com/liammlrb-eng/nevada-county-experiences
Operator day-to-day guide	operator_guide.pdf (in this bundle)
Long-form pitch & UX rationale	demo_pitch.pdf (in this bundle)
Slide deck	demo_pitch.pptx (in this bundle)

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